CAREER POINT

JEE Main Online Exam 2020

Questions & Solutions

9th January 2020 | Shift - II

PHYSICS

Q.1 For the four sets of three measured physical quantities as given below. Which of the following options is correct ?

(i) $A_1 = 24.36, B_1 = 0.0724, C_1 = 256.2$ (ii) $A_2 = 24.44, B_2 = 16.082, C_2 = 240.2$

(iii) $A_3 = 25.2, B_3 = 19.2812, C_3 = 236.183$ (iv) $A_4 = 25, B_4 = 236.191, C_4 = 19.5$

(1) $A_4 + B_4 + C_4 < A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3$

(2) $A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3 = A_4 + B_4 + C_4$

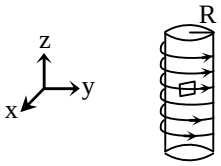
(3) $A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2 < A_4 + B_4 + C_4$

(4) $A_4 + B_4 + C_4 < A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2$

Ans. **[Bonus] [NTA Answer Key (1)]**

Sol. No option is matching.

Q.2 An electron gun is placed inside a long solenoid of radius R on its axis. The solenoid has a turns/length and caries a current I. The electron gun shoots an electron along the radius of the solenoid with speed ν. If the electron does not hit the surface of the solenoid, maximum possible value of ν is (all symbols have their standard meaning)



(1) $\frac{e\mu_0 nIR}{4m}$ (2) $\frac{e\mu_0 nIR}{2m}$ (3) $\frac{e\mu_0 nIR}{m}$ (4) $\frac{2e\mu_0 nIR}{m}$

Ans. **[2]**

Sol. $$\frac{R}{2} = \frac{mV}{qB}$$

$$\Rightarrow \quad \frac{R}{2} = \frac{mV}{q\mu_0 nI}$$

$$\Rightarrow \quad V = \frac{e\mu_0 nIR}{2m}$$

Q.3 A spring mass system (mass m, spring constant k and natural length ℓ) rests in equilibrium on a horizontal disc. The free end of the spring is fixed at the centre of the disc. If the disc together with spring mass system, rotated about it's axis with an angular velocity ω, ($k > m\omega^2$) the relative change in the length of the spring is best given by the option -

(1) $\frac{m\omega^2}{3k}$ (2) $\frac{2m\omega^2}{k}$ (3) $\frac{m\omega^2}{k}$ (4) $\sqrt{\frac{2}{3}}\left(\frac{m\omega^2}{k}\right)$

Ans. **[3]**

Sol. $m\omega^2(\ell + x) = kx$

$\Rightarrow \quad 1 + \frac{\ell}{x} = \frac{k}{m\omega^2}$

$\Rightarrow \quad \frac{\ell}{x} = \frac{k}{m\omega^2} - 1$

As $\quad \frac{k}{m\omega^2} >> 1$

$\therefore \quad \frac{\ell}{x} = \frac{k}{m\omega^2}$

$\Rightarrow \quad \left[\frac{x}{\ell} = \frac{m\omega^2}{k}\right]$

Q.4 A particle starts from the origin at t = 0 with an initial velocity of $3.0\,\hat{i}$ m/s and moves in the x-y plane with a constant acceleration $(6.0\,\hat{i} + 4.0\,\hat{j})$ m/s^2. The x-coordinate of the particle at the instant when its y-coordinate is 32 m is D metres. The value of D is -

(1) 50 (2) 60 (3) 40 (4) 32

Ans. **[2]**

Sol. $\vec{u} = 3\hat{i}$

$\vec{a} = 6\hat{i} + 4\hat{j}$

$\vec{S} = \vec{u}\,t + \frac{1}{2}\vec{a}\,t^2$

$x = 3t + \frac{1}{2} \times 6\,t^2$

$= 3t + 3t^2 \qquad \ldots. (1)$

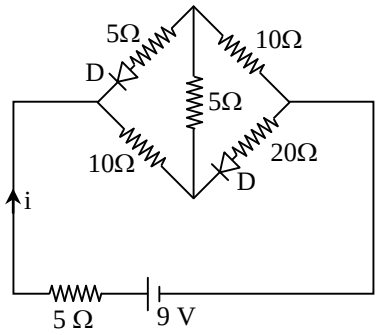
$y = \frac{1}{2} \times 4 \times t^2 = 2t^2 = 32$ (Given)

$\therefore \quad t = 4$ sec $\qquad \ldots.(2)$

From (1) & (2) :

$x = 3 \times 4 + 3 \times 4^2 = 12 + 48 = 60$ m

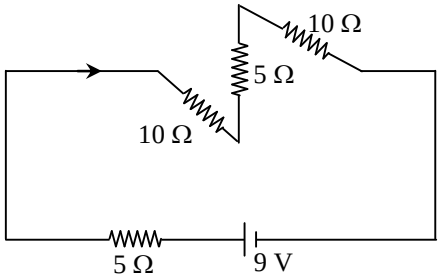
Q.5 The current I in the network is :



(1) 0.3 A (2) 0 A (3) 0.6 A (4) 0.2 A

Ans. **[1]**

Sol. Both the diodes are reverse-biased and hence they can be considered as open circuit.



$i = \frac{9}{30} = 0.3$ A

Q.6 Two gases-argon (atomic radius 0.07 nm, atomic weight 40) and xenon (atomic radius 0.1 nm, atomic weight 140) have the same number density and are at the same temperature. The ratio of their respective mean free times is closest to -

(1) 2.3 (2) 4.67 (3) 1.83 (4) 3.67

Ans. **[3] [NTA Answer Key (4) 3.67]**

Sol. Mean free time $= \frac{1}{\sqrt{2}n\pi V_{avg}d^2}$

Where, $V_{arg} = \sqrt{\frac{8RT}{\pi M}}$

$T \propto \frac{1}{V_{avg}d^2}$

$\propto \sqrt{\frac{M}{d^2}}$

$\frac{\tau_{Ar}}{\tau_{Xe}} = \sqrt{\frac{M_{Ar}}{M_{xe}}} \cdot \frac{d_{xe}^2}{d_{Ar}^2}$

$= \sqrt{\frac{40}{140}} \times \left(\frac{0.10}{0.07}\right)^2$

$= 0.534 \times 2.04 = 1.089$

$= 1.089$

Closest to 1.83

Q.7 An electron of mass m and magnitude of charge |e| initially at rest gets accelerated by a constant electric field E. The rate of change of de-Broglie wavelength of this electron at time t ignoring relativistic effects is -

(1) $\frac{-h}{|e|Et^2}$ (2) $-\frac{h}{|e|E\sqrt{t}}$ (3) $\frac{|e|Et}{h}$ (4) $-\frac{h}{|e|Et}$

Ans. **[1]**

Sol. $F = |e|E$

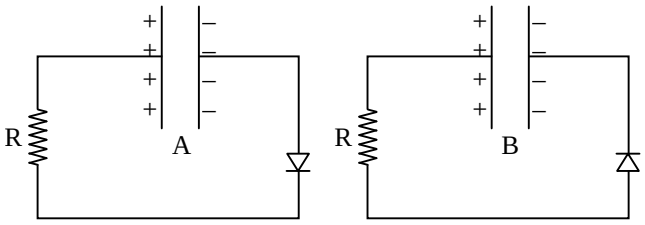
$a = \frac{F}{m} = \frac{|e|E}{m}$

$V = at = \frac{|e|E}{m}t$

$\lambda = \frac{h}{mV} = \frac{h}{|e|Et}$

$\frac{d\lambda}{dt} = \frac{-h}{|e|Et^2}$

Q.8 Two identical capacitors A and B, charged to the same potential 5V are connected in two different circuits as shown below at time t = 0. If the charge on capacitors A and B at time t = CR is Q_A and Q_B respectively, then (Here e is the base of natural logarithm)

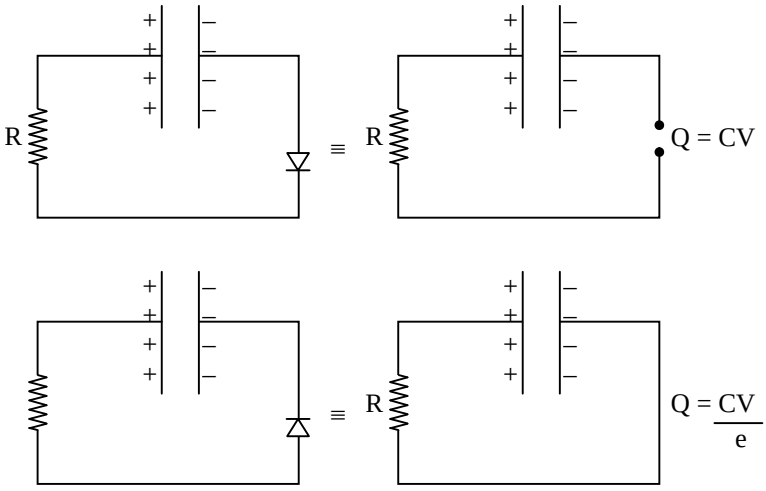


(1) $Q_A = VC$, $Q_B = \frac{VC}{e}$ (2) $Q_A = \frac{CV}{2}$, $Q_B = \frac{VC}{e}$

(3) $Q_A = VC$, $Q_B = CV$ (4) $Q_A = \frac{VC}{e}$, $Q_B = \frac{CV}{2}$

Ans. **[1]**

Sol.



Q.9 A wire of length L and mass per unit length 6.0×10^{-3} kg m^{-1} is put under tension of 540 N. Two consecutive frequencies that it resonates at are : 420 Hz and 490 Hz. Then L in meters is -

(1) 2.1 m (2) 1.1 m (3) 8.1 m (4) 5.1 m

Ans. **[1]**

Sol. $V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{540}{6\times10^{-3}}} = 300$ m/s

$\frac{V}{2L} = 490–420$

$\Rightarrow \quad L = \frac{V}{2\times70} = \frac{300}{2\times70} = 2.1$ m

Q.10 The energy required to ionize a hydrogen like ion in its ground state is 9 Rydbergs. What is the wavelength of the radiation emitted when the electron in this ion jumps from the second excited state to the ground state ?

(1) 11.4 nm (2) 24.2 nm (3) 8.6 nm (4) 35.8 nm

Ans. **[1]**

Sol. $E_n = 13.6 \frac{z^2}{n^2}$ eV

For ground state : n = 1

$E_1 = -13.6\, z^2 = -13.6 \times 9$

$\boxed{Z = 3}$

Energy of photon emitted

$= 13.6z^2 \left(\frac{1}{1^2} - \frac{1}{3^2}\right)$

$= 13.6 \times 3^2 \times \frac{8}{9}$

$= 108.8$ eV

$\therefore \quad \lambda = \frac{1240}{108.8}$ nm $= 11.39$ nm ≈ 11.4 nm

Q.11 In LC circuit the inductance L = 40 mH and capacitance C = 100 μF. If a voltage V(t) = 10 sin (314 t) is applied to the circuit, the current in the circuit is given as -

(1) 10 cos 314 t (2) 0.52 sin 314 t (3) 5.2 cos 314 t (4) 0.52 cos 314 t

Ans. **[4]**

Sol. L = 40 mH

C = 100 μF

w = 314

$X_c = \frac{1}{\omega c} = \frac{1}{314\times100\times10^{-6}}$

$= \frac{10000}{314} = 31.85\ \Omega$

$X_L = \omega L = 314 \times 40 \times 10^{-3} = 12.56\ \Omega$

$I_0 = \frac{V_0}{X_c - X_L} = \frac{10}{19.29} = 0.52$

$\therefore \quad I = 0.52 \sin(314t + \pi/2)$

$= 0.52 \cos(314\ t)$

Q.12 A plane electromagnetic wave is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$, with its polarization along the direction $\hat{k}$. The correct form of the magnetic field of the wave would be (here B_0 is an appropriate constant)

(1) $B_0\,\hat{k}\,\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(2) $B_0\,\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t + k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(3) $B_0\,\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(4) $B_0\,\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

Ans. **[4]**

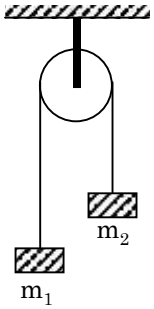
Sol. Wave propagates in the direction of $\vec{E}\times\vec{B}$.

Given : $\vec{E}$ is along $\hat{k}$ and wave propagates along $\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

Check : $\hat{B} = \frac{\hat{i}-\hat{j}}{\sqrt{2}}$

$\hat{E}\times\hat{B} = \hat{k}\times\left(\frac{\hat{i}-\hat{j}}{\sqrt{2}}\right) = \frac{\hat{i}+\hat{j}}{\sqrt{2}}$

Q.13 A uniformly thick wheel with moment of inertia I and radius R is free to rotate about its centre of mass (see fig.). A massless string is wrapped over its rim and two blocks of masses m_1 and m_2 ($m_1 > m_2$) are attached to the ends of the string. The system is released from rest. The angular speed of the wheel when m_1 descents by a distance h is -



(1) $\left[\frac{2(m_1-m_2)gh}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}$

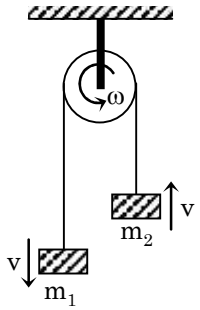
(2) $\left[\frac{2(m_1+m_2)gh}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}$

(3) $\left[\frac{m_1+m_2}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}gh$

(4) $\left[\frac{(m_1-m_2)}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}gh$

Ans. **[1]**

Sol.



By conservation of energy :

Loss in energy = Gain in enrgy

$m_1gh = m_2gh + \frac{1}{2} m_1V^2 + \frac{1}{2} m_2V^2 + \frac{1}{2} I\omega^2$

$(m_1 - m_2) gh = \frac{1}{2} (m_1 + m_2) V^2 + \frac{1}{2} I\omega^2$

$= \frac{1}{2}(m_1 + m_2)R^2\omega^2 + \frac{1}{2} I\omega^2$

$= \frac{1}{2} [I + (m_1 + m_2) R^2] \omega^2$

$\omega^2 = \frac{2(m_1 - m_2)gh}{I + (m_1 + m_2)R^2}$

$\Rightarrow \omega = \left[\frac{2(m_1 - m_2)gh}{I + (m_1 + m_2)R^2}\right]^{1/2}$

Q.14 A small circular loop of conducting wire has radius a and carries current I. It is placed in a uniform magnetic field B perpendicular to its plane such that when rotated slightly about its diameter and released, it starts performing simple harmonic motion of time period T. If the mass of the loop is m then -

(1) $T = \sqrt{\frac{2\pi m}{IB}}$ (2) $T = \sqrt{\frac{2m}{IB}}$ (3) $T = \sqrt{\frac{\pi m}{2IB}}$ (4) $T = \sqrt{\frac{\pi m}{IB}}$

Ans. **[1]**

Sol. $\mu = I \cdot \pi a^2$

$\tau = \mu B \sin\theta$

$= I \cdot \pi a^2 B \sin\theta$

$\approx (I \pi a^2 B) \theta$

Moment of inertia, $I' = \frac{ma^2}{2}$

$\therefore \quad T = 2\pi\sqrt{\frac{ma^2}{2I\pi a^2 B}}$

$T = \sqrt{\frac{2m\pi}{IB}}$

Q.15 A particle of mass m is projected with a speed u from the ground at an angle $\theta = \frac{\pi}{3}$ w.r.t. horizontal (x-axis). When it has reached its maximum height, it collides completely inelastically with another particle of the same mass and velocity $u\hat{i}$. The horizontal distance covered by the combined mass before reaching the ground is -

(1) $\frac{3\sqrt{3}}{8}\frac{u^2}{g}$ (2) $2\sqrt{2}\frac{u^2}{g}$ (3) $\frac{5}{8}\frac{u^2}{g}$ (4) $\frac{3\sqrt{2}}{4}\frac{u^2}{g}$

Ans. **[1]**

Sol. $mu + mu\cos\theta = 2mV$

$$V = \frac{u + u\cos\theta}{2}$$

$$= \frac{u + u \times \frac{1}{2}}{2}$$

$$= \frac{3u}{4}$$

$$H_{max} = \frac{u^2 \sin^2 60^\circ}{2g} = \frac{u^2 \times \frac{3}{4}}{2g} = \frac{3u^2}{8g}$$

$$\text{Time taken} = \sqrt{\frac{2H_{max}}{g}} = \sqrt{\frac{2}{g} \times \frac{3u^2}{8g}} = \frac{\sqrt{3}}{2}\ \frac{u}{g}$$

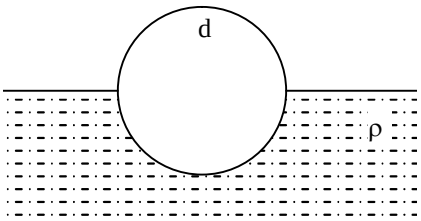
$$\text{Horizontal distance traveled} = \frac{3u}{4} \cdot \frac{\sqrt{3}}{2}\ \frac{u}{g} = \frac{3\sqrt{3}}{8}\ \frac{u^2}{g}$$

Q.16 A small spherical droplet of density d is floating exactly half immersed in a liquid of density ρ and surface tension T. The radius of the droplet is (take note that the surface tension applies an upward force on the droplet)

(1) $r = \sqrt{\frac{T}{(d-\rho)g}}$ (2) $r = \sqrt{\frac{3T}{(2d-\rho)g}}$ (3) $r = \sqrt{\frac{T}{(d+\rho)g}}$ (4) $r = \sqrt{\frac{2T}{3(d+\rho)g}}$

Ans. **[2]**

Sol.



$$T \cdot 2\pi R + \frac{2}{3}\pi R^3 \rho g = \frac{4}{3}\pi R^3 dg$$

$$\Rightarrow \quad T + \frac{R^2\rho g}{3} = \frac{2R^2 dg}{3}$$

$$\Rightarrow \quad T = \frac{R^2 g}{3}(2d-\rho)$$

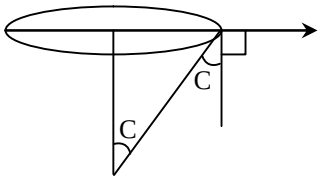
$$\Rightarrow \quad R = \sqrt{\frac{3T}{g(2d-\rho)}}$$

Q.17 There is a small sourceof light at some depth below the surface of water (refractive index = $\frac{4}{3}$) in a tank of large cross sectional surface area. Neglecting any reflection from the bottom and absorption by water, percentage of light that emerges out of surface is (nearly) : [Use the fact that surface area of a spherical cap of height h and radius of curvature r is $2\pi rh$]

(1) 21 % (2) 34 % (3) 50 % (4) 17 %

Ans. **[4]**

Sol.



$\sin C = \frac{3}{4}$

$\cos C = \sqrt{1-\frac{9}{16}} = \frac{\sqrt{7}}{4}$

Surface area in solid angle $d\Omega = 2\pi R^2 (1 - \cos C) = 2\pi R^2 \left(1-\frac{\sqrt{7}}{4}\right)$

Percentage of light = $\frac{2\pi R^2}{4\pi R^2}\left(1-\frac{\sqrt{7}}{4}\right) \times 100\ \%$

$= \frac{4-\sqrt{7}}{8} \times 100\ \%$

$= 17\ \%$

Q.18 Planet A has mass M and radius R. Planet B has half the mass and half the radius of planet A. if the escape velocities from the Planets A and B are v_A and v_B, respectively, then $\frac{v_A}{v_B} = \frac{n}{4}$. The value of n is -

(1) 2 (2) 3 (3) 1 (4) 4

Ans. **[4]**

Sol. $V_0 = \sqrt{\frac{2GM}{R}}$

$V_A = \sqrt{\frac{2GM}{R}}$

$V_B = \sqrt{\frac{2G\frac{M}{2}}{\frac{R}{2}}} = \sqrt{\frac{2GM}{R}}$

$\frac{V_A}{V_B} = 1$

$\Rightarrow \quad \frac{n}{4} = 1$

$\Rightarrow \quad n = 4$

Q.19 A rod of length L has non-uniform linear mass density given by $\rho(x) = a + b\left(\frac{x}{L}\right)^2$, where a and b are constants and $0 \le x \le L$. The value of x for the centre of mass of the rod is at -

(1) $\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)L$ (2) $\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)L$ (3) $\frac{3}{2}\left(\frac{a+b}{2a+b}\right)L$ (4) $\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)L$

Ans. **[4]**

Sol. $\rho = a + b\left(\frac{x}{L}\right)^2$

$dm = \rho dx = \left(a + b\frac{x^2}{L^2}\right)dx$

$M = \int dm = \int_0^L \left(a + \frac{bx^2}{L^2}\right)dx$

$X_{com} = \frac{\int x\,dm}{\int dm} = \frac{3\ell}{4}\left(\frac{2a+b}{3a+b}\right)$

Q.20 Two steel wires having same length are suspended from a ceiling under the same load. If the ratio of their energy stored per unit volume is 1 : 4, the ratio of their diameters is -

(1) 2 : 1 (2) $\sqrt{2}:1$ (3) $1:\sqrt{2}$ (4) 1 : 2

Ans. **[2]**

Sol. $u = \frac{1}{2} \times \text{stress} \times \text{strain}$

$= \frac{(\text{stress})^2}{2y}$

$= \frac{(F/A)^2}{2y} \propto \frac{1}{A^2}$

$\frac{u_1}{u_2} = \frac{A_2^2}{A_1^2} = \frac{1}{4}$

$\Rightarrow A_1 = 2A_2$

$\Rightarrow d_1^2 = 2d_2^2$

$\Rightarrow \left[\frac{d_1}{d_2} = \sqrt{2}:1\right]$

Q.21 In a Young's double slit experiment 15 fringes are observed on a small portion of the screen when light of wavelength 500 nm is used. Ten fringes are observed on the same section of the screen when another light source of wavelength λ is used. Then the value of λ is (in nm).

Ans. **[750]**

Sol. $15 \times 500\,\frac{D}{d} = 10 \times \lambda\,\frac{D}{d}$

$\lambda = \frac{15\times 500}{10} = 750$ nm

Q.22 Starting at temperature 300 K, one mole of an ideal diatomic gas ($\gamma = 1.4$) is first compressed adiabatically from volume V_1 to $V_2 = \frac{V_1}{16}$. It is then allowed to expand isobarically to volume $2V_2$. If all the processes are the quasi-static then the final temperature of the gas (in ^oK) is (to the nearest integer) …….. .

Ans. **[1819] [NTA Answer : 1818]**

Sol. $T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

$$T_2 = 300\left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$= 300\,(16)^{0.4}$$

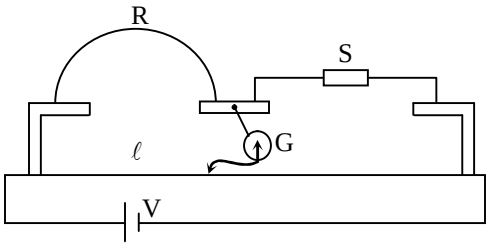
$$T_{final} = \frac{P_2 \times 2V_2}{nR} = 2\left(\frac{P_2V_2}{nR}\right)$$

$$= 2T_2$$

$$= 600 \times (16)^{0.4}$$

$$= 1819\ K$$

Q.23 In a meter bridge experiment S is a standard resistance. R is a reistance wire. It is found that balancing length is ℓ = 25 cm. If R is replaced by a wire of half length and half diameter that of R of same material, then the balancing distance ℓ' (in cm) will now be …….. .



Ans. **[40.00]**

Sol. $\frac{X}{R} = \frac{75}{25} = 3$

$$R = \frac{\rho L}{A} = \frac{4\rho L}{\pi d^2}$$

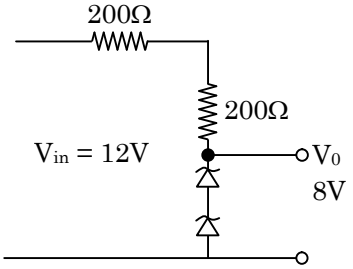
$$R' = \frac{4\rho\left(\frac{\ell}{2}\right)}{\pi\left(\frac{d}{2}\right)^2} = 2R$$

Then, $\frac{X}{R'} = \frac{100-\ell}{\ell}$

$\Rightarrow \quad \frac{100-\ell}{\ell} = \frac{X}{2R} = \frac{3}{2}$

$\Rightarrow \quad [\ell = 40\ cm]$

Q.24 The circuit shown below is working as a 8V dc regulated voltage source. When 12 V is used as input, the power dissipated (in mW) in each diode is : (considering both zener diodes are identical) …… .

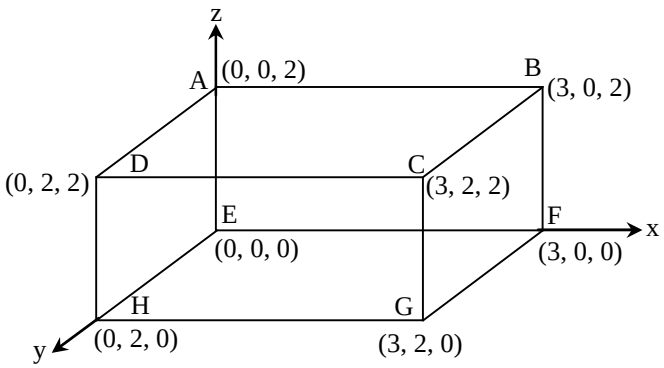


Ans. **[40.00] [NTA Answer : 12.00]**

Sol. $I = \frac{12-8}{200+200} = 10^{-2}\ A$

$P = 4 \times 10^{-2} = 40\ mW$

Q.25 An electric field $\vec{E} = 4x\hat{i} - (y^2 + 1)\hat{j}$ N/C passes through the box shown in figure. The flux of the electric field through surfaces ABCD and BCGF are marked as ϕ_1 and ϕ_{11} respectively. The difference between $(\phi_1 - \phi_{11})$ is (in Nm^2/C) …….. .



Ans. **[–48] [NTA Answer : 48]**

Sol. $\vec{E} = 4x\hat{i} - (y^2 + 1)\hat{j}$

$\phi_1 = \int \vec{E}.\overrightarrow{dA} = 0$

$\phi_2 = \int \vec{E}.\overrightarrow{dA} = \vec{E}.\vec{A} = 16\ x = 16 \times 3 = 48$

$\phi_I - \phi_{II} = 0 - 48 = -48\ \frac{N - m^2}{c^2}$

JEE Main Online Exam 2020

Questions & Solutions

9th January 2020 | Shift - II

CHEMISTRY

Q.1 Amongst the following, the form of water with the lowest ionic conductance at 298 K is :

(1) sea water (2) distilled water

(3) saline water used for intravenous injection (4) water from a well

Ans. **[2]**

Sol. The form of H_2O with the lowest ionic conductance at 298 K is distilled water.

Q.2 The correct order of the spin-only magnetic moments of the following complexes is :

(I) $[Cr(H_2O)_6]Br_2$ (II) $Na_4[Fe(CN)_6]$

(III) $Na_3[Fe(C_2O_4)_3](\Delta_0 > P)$ (IV) $(Et_4N)_2\,[CoCl_4]$

(1) (I) > (IV) > (III) > (II) (2) (II) ≈ (I) > (IV) > (III)

(3) (III) > (I) > (IV) > (II) (4) (III) > (I) > (II) > (IV)

Ans. **[1]**

Sol. $[Cr(H_2O)_6]Br_2$: $Cr^{2+} = [Ar]\ 4s^0\ 3d^4$

$$t_{2g}^3\ eg^1$$

$$n = 4,\ \mu = \sqrt{n(n+2)} = \sqrt{24}\ BM$$

$Na_4[Fe(CN)_6]$: $Fe^{2+} = [Ar]\ 4s^0\ 3d^6$

$$t_{2g}^6\ eg^0$$

$$n = 0,\ \mu = 0$$

$Na_3[Fe(C_2O_4)_3]$: $Fe^{3+} = [Ar]\ 3d^5$

$$t_{2g}^5\ eg^0$$

$$n = 1,\ \mu = \sqrt{3}\ BM$$

$(Et_4N)_2\,[CoCl_4]$: $Co^{2+} = [Ar]\ 3d^7$

$$e_g^4\ t_{2g}^3$$

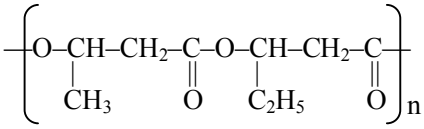
$$n = 3,\ \mu = \sqrt{15}\ BM$$

Q.3 Which polymer has 'chiral' monomer(s) ?

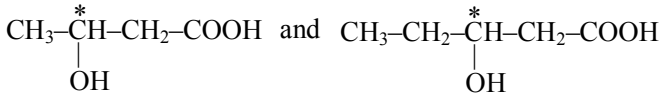
(1) Nylon 6, 6 (2) Neoprene (3) PHBV (4) Buna-N

Ans. **[3]**

Sol. PHBV :



Monomer :



3-Hydroxybutanoic acid 3-Hydroxypentanoic acid

Q.4 The first and second ionisation enthalpies of a metal are 496 and 4560 kJ mol^{-1}, respectively. How many moles of HCl and H_2SO_4, respectively, will be needed to react completely with 1 mole of the metal hydroxide ?

(1) 1 and 1 (2) 1 and 2 (3) 2 and 0.5 (4) 1 and 0.5

Ans. **[4]**

Sol. Metal : First ionization enthalpies = 496 kJ/mole
Second ionization enthalpies = 4560 kJ/mol
According to the given information, ionization enthalpies Metal belong to 1st group i.e. Monovalent cation

$$\underset{1\text{ mole}}{MOH} + \underset{1\text{ mole}}{HCl} \rightarrow MCl + H_2O$$

$$\underset{1\text{ mole}}{2MOH} + \underset{1/2\text{ mole}}{H_2SO_4} \longrightarrow M_2SO_4 + H_2O$$

Q.5 Among the statements (a) – (d), the correct ones are :

(a) Lithium has the highest hydration enthalpy among the alkali metals.
(b) Lithium chloride is insoluble in pyridine.
(c) Lithium cannot form ethynide upon its reaction with ethyne
(d) Both lithium and magnesium react slowly with H_2O

(1) (a), (b) and (d) only (2) (a) and (d) only
(3) (b) and (c) only (4) (a), (c) and (d) only

Ans. **[4]**

Sol. *Lithium has the highest hydration enthalpy among the alkali metals due to small size.

*Lithium chloride is covalent in nature so it's soluble in non-polar solvent.

*Lithium and Magnesium react slowly with H_2O.

Q.6 The solubility product of $Cr(OH)_3$ at 298 K is 6.0×10^{-31}. The concentration of hydroxide ions in a saturated solution of $Cr(OH)_3$ will be :

(1) $(18 \times 10^{-31})^{1/4}$ (2) $(4.86 \times 10^{-29})^{1/4}$ (3) $(18 \times 10^{-31})^{1/2}$ (4) $(2.22 \times 10^{-31})^{1/4}$

Ans. **[1]**

Sol. $Cr(OH)_3 \rightleftharpoons \underset{s}{Cr^{3+}} + \underset{3s}{3OH^-}$

$K_{sp} = [Cr^{3+}]\,[OH^-]^3$

$6 \times 10^{-31} = S \times (3S)^3$

$6 \times 10^{-31} = 27\,S^4$

$$S = \left(\frac{6}{27} \times 10^{-31}\right)^{1/4}$$

$[OH^-] = 3S$

$$= 3\left(\frac{6}{27} \times 10^{-31}\right)^{1/4} = (18 \times 10^{-31})^{1/4}\ M$$

Q.7 5 g of zinc is treated separately with an excess of
(a) dilute hydrochloric acid and
(b) aqueous sodium hydroxide.
The ratio of the volumes of H_2 evolved in these two reactions is : -

(1) 1 : 4 (2) 1 : 1 (3) 1 : 2 (4) 2 : 1

Ans. **[2]**

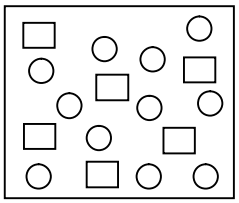
Sol. $Zn + 2dil.\ HCl \longrightarrow ZnCl_2 + H_2$

$Zn + 2NaOH \longrightarrow Na_2ZnO_2 + H_2$

$$\text{Mole of Zn} = \frac{\text{Mole of dil HCl}}{2} = \frac{\text{Mole of NaOH}}{2}$$

$$\frac{\text{volume of HCl}}{\text{volume of NaOH}} = \frac{1}{1}$$

Q.8 If the figure shown below reactant A (represented by square) is in equilibrium with product B (represented by circle). The equilibrium constant is :



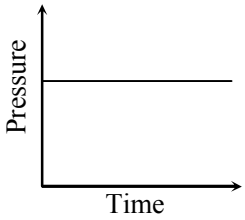
(1) 8 (2) 2 (3) 4 (4) 1

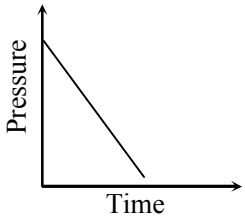
Ans. **[2]**

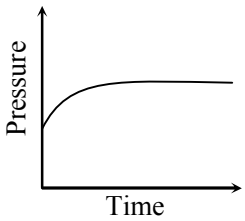
Sol. $A \longrightarrow B$

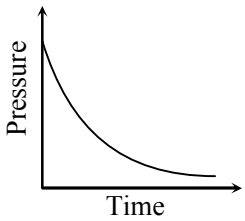
$$K_{eq.} = \frac{[B]}{[A]} = \frac{11}{6}$$

Q.9 A mixture of gases O_2, H_2 and CO are taken in a closed vessel containing charcoal. The graph that represents the correct behaviour of pressure with time is :

(1)


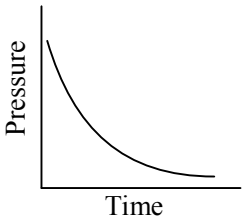
(2)


(3)


(4)


Ans. **[4]**

Sol.

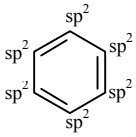


on increasing time, pressure will be decreases.

Q.10 The number of sp^2 hybrid orbitals in a molecule of benzene is -

(1) 18 (2) 6 (3) 12 (4) 24

Ans. **[1]**

Sol. Benzene : C_6H_6 

In Benzene, Each carbon is sp^2 Hybridize

Total number of carbon = 6(sp^2 Hybri.)

$\therefore$ Total Hybrid orbital = $6 \times 3 = 18$

Q.11 Biochemical Oxygen Demand (BOD) is the amount of oxygen required (in ppm) :

(1) for the photochemical breakdown of waste present in 1 m^3 volume of a water body.

(2) for sustaining life in a water body.

(3) by bacteria to break-down organic waste in a certain volume of a water sample.

(4) by anaerobic bacteria to breakdown inorganic waste present in a water body.

Ans. **[3]**

Sol. Biochemical oxygen demand (BOD)

The amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water.

Q.12 The true statement amongst the following is :

(1) S is a function of temperature but ΔS is not a function of temperature.

(2) Both ΔS and S are functions of temperature.

(3) S is not a function of temperature but ΔS is a function of temperature.

(4) Both S and ΔS are not functions of temperature.

Ans. **[2]**

Sol. $S = \int \frac{dq}{T}$

$\Delta S = nC \int_{T_1}^{T_2} dT$

Both ΔS and S are function of temperature.

Q.13 A, B and C are three biomolecules. The results of the tests performed on them are given below :

	Molisch's Test	Barfoed Test	Biuret Test
A	Positive	Negative	Negative
B	Positive	Positive	Negative
C	Negative	Negative	Positive

A, B and C are respectively :

(1) A = Lactose, B = Glucose, C = Alanine

(2) A = Lactose, B = Glucose, C = Albumin

(3) A = Glucose, B = Fructose, C = Albumin

(4) A = Lactose, B = Fructose, C = Alanine

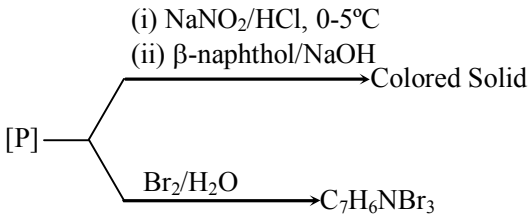
Ans. **[2]**

Sol. Lactose : Molisch's Test

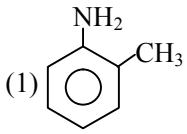
Glucose : Molisch's Test and Barfoed Test

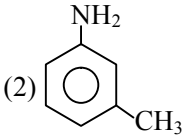
Alumin : Biuret Test

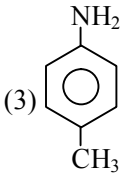
Q.14 Consider the following reactions,

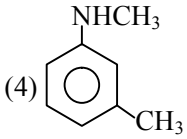


The compound [P] is :

(1) 

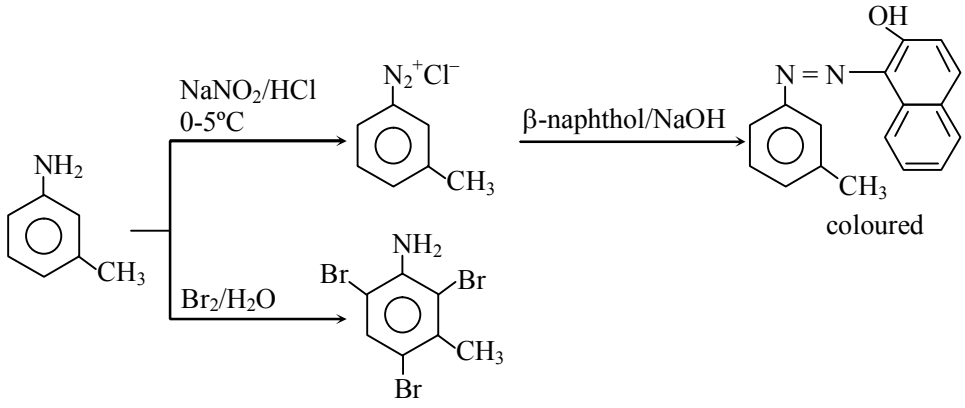
(2) 

(3) 

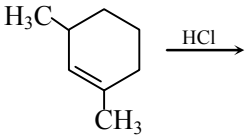
(4) 

Ans. **[2]**

Sol.

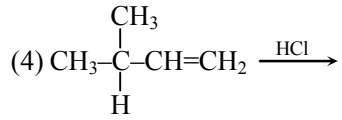


Q.15 Which of the following reactions will not produce a racemic product ?

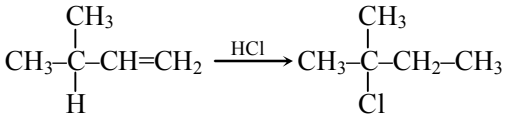
(1) 

(2) $CH_3CH_2CH{=}CH_2 \xrightarrow{HBr}$

(3) 

(4) 

Ans. **[4]**

Sol. 

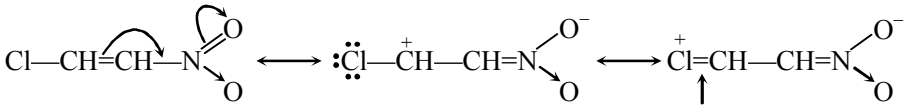
optical inactive

Q.16 Which of the following has the shortest C–Cl bond ?

(1) $Cl{-}CH{=}CH{-}NO_2$ (2) $Cl{-}CH{=}CH{-}CH_3$

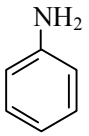
(3) $Cl{-}CH{=}CH{-}OCH_3$ (4) $Cl{-}CH{=}CH_2$

Ans. **[1]**

Sol. 

double bond char.

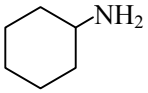
Q.17 The decreasing order of basicity of the following amines is :



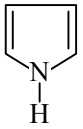
(I)



(II)



(III)

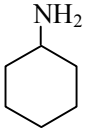


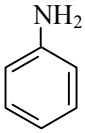
(IV)

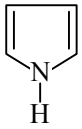
(1) (III) > (I) > (II) > (IV) (2) (II) > (III) > (IV) > (I)

(3) (I) > (III) > (IV) > (II) (4) (III) > (II) > (I) > (IV)

Ans. **[4]**

Sol.  : Nitrogen ℓ.p. not participate in resonance.

 : Nitrogen ℓ.p. participate in resonance.

 : Nitrogen ℓ.p. participate in resonance and increase the stability of the compound due to aromicity.

 : Nitrogen ℓ.p. not participate in resonance due to increase the stability of the compound.

Q.18 The reaction of $H_3N_3B_3Cl_3$(A) with $LiBH_4$ in tetrahydrofuran gives inorganic benzene (B). Further, the reaction of (A) with (C) leads to $H_3N_3B_3(Me)_3$. Compounds (B) and (C) respectively, are :

(1) Borazine and MeBr (2) Boron nitride and MeBr

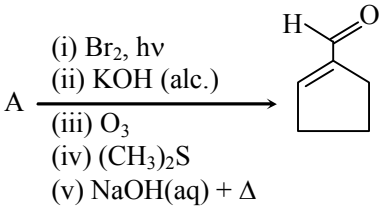
(3) Diborane and MeMgBr (4) Borazine and MeMgBr

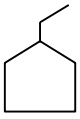
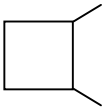
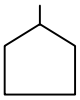
Ans. **[4]**

Sol. $\underset{(A)}{B_3N_3H_3Cl_3} + LiBH_4 \longrightarrow \underset{\substack{(B)\\ \text{Inorganic Benzene or Borazine}}}{B_3N_3H_6}$

$B_3N_3H_3Cl + \underset{(C)}{MeMgBr} \longrightarrow B_3N_3H_3(CH_3)_3 + 3MgBrCl$

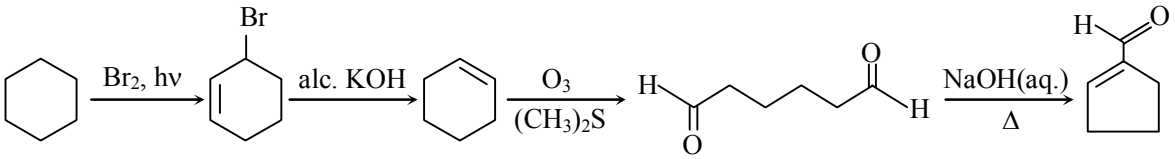
Q.19 In the following reaction A is :



(1)  (2)  (3)  (4) 

Ans. **[2]**

Sol.

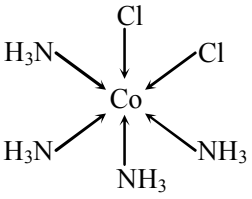


Q.20 The isomer (s) of $[Co(NH_3)_4Cl_2]$ that has/have a Cl–Co–Cl angle of 90º, is/are :

(1) meridional and trans (2) cis only (3) trans only (4) cis and trans

Ans. **[2]**

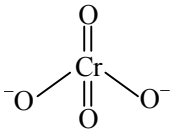
Sol. $[Co(NH_3)_4Cl_2]$



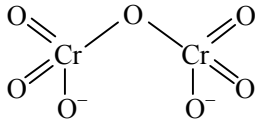
Q.21 The sum of the total number of bonds between chromium and oxygen atoms in chromate and dichromate ions is _____

Ans. **[12]**

Sol. Chromate = CrO_4^{2-}



dichromate = $Cr_2O_7^{2-}$



Total number of Cr–O bond = 12

Q.22 A sample of milk splits after 60 min. at 300 K and after 40 min. at 400 K when the population of *lactobacillus acidophilus* in it doubles. The activation energy (in kJ/mol) for this process is closest to _____

(Given, R = 8.3 J $mol^{-1}K^{-1}$, $\ell n\left(\frac{2}{3}\right) = 0.4$, $e^{-3} = 4.0$)

Ans. **[3.98]**

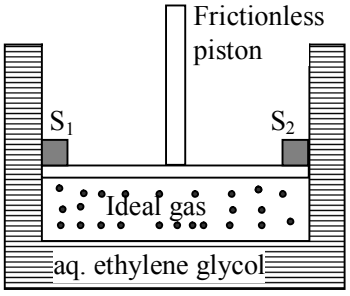
Sol. $\ell n\frac{K_2}{K_1} = \frac{Ea}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

$\ell n\frac{60}{40} = \frac{Ea}{8.3}\left[\frac{1}{300} - \frac{1}{400}\right]$

Ea = 3.98 kJ/mole.

Q.23 A Cylinder containing an ideal gas (0.1 mol of 1.0 dm^3) is in thermal equilibrium with a large volume of 0.5 molal aqueous solution of ethylene glycol at its freezing point. If the stoppers S_1 and S_2 (as shown in the figure) are suddenly withdrawn, the volume of the gas in litres after equilibrium is achieved will be _____.

(Given, K_f (water) = 2.0 K kg mol^{-1}, R = 0.08 dm^3 atm K^{-1} mol^{-1})



Ans. **[2.18]**

Sol. $\Delta T_f = K_f \times i \times m$

$\Delta T_f = 2.0 \times 1 \times 0.5$

$\Delta T_f = 1$

$273 - T_i = 1$

$T_i = 272$ K

$P = \frac{nRT}{V}$

$P = \frac{0.1 \times 0.08 \times 272}{1}$

P = 2.176 atm

Apply Boyle's law

$P_1V_1 = P_2V_2$

$2.176 \times 1 = 1 \times V_2$

$V_2 = 2.17$

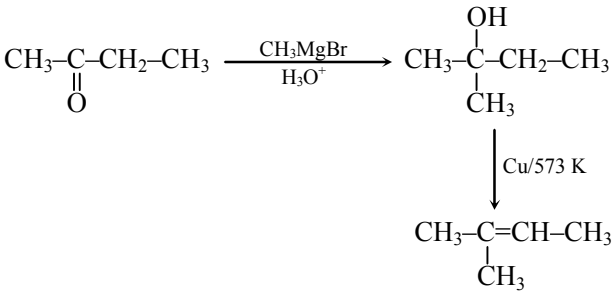
Q.24 Consider the following reactions

$$A \xrightarrow[(ii)\,H_3O^+]{(i)\,CH_3MgBr} B \xrightarrow[573\,K]{Cu} \text{2-methyl-2-butene}$$

The mass percentage of carbon in A is _____.

Ans. **[66.67]**

Sol.



$$\%\ \text{carbon} = \frac{\text{Atomic mass} \times \text{Atomicity}}{\text{Molar mass}} \times 100$$

$$= \frac{12 \times 4}{72} \times 100 = 66.66\%$$

Q.25 10.30 mg of O_2 is dissolved into a liter of sea water of density 1.03 g/mL. The concentration of O_2 in ppm is _____.

Ans. **[10]**

Sol. $$\text{PPM} = \frac{\text{Mass of } O_2}{\text{Mass of water}} \times 10^6$$

$$= \frac{10.30 \times 10^{-3}}{1030} \times 10^6 = 10$$

JEE Main Online Exam 2020

Questions & Answer

9th January 2020 | Shift - II

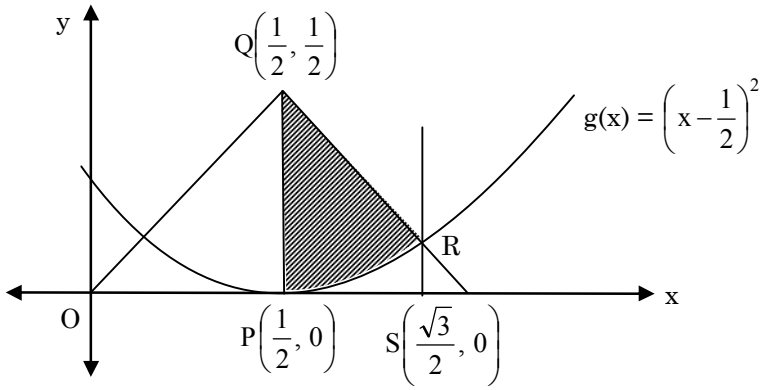
MATHEMATICS

Q.1 Given : $f(x) = \begin{cases} x & ; \ 0 \le x < \frac{1}{2} \\ \frac{1}{2} & ; \ x = \frac{1}{2} \\ 1-x & ; \ \frac{1}{2} < x \le 1 \end{cases}$ and $g(x) = \left(x - \frac{1}{2}\right)^2$, $x \in R$. Then the area (in sq. units) of the region bounded by the curves $y = f(x)$ and $y = g(x)$ between the lines, $2x = 1$ and $2x = \sqrt{3}$, is -

(1) $\frac{1}{3} + \frac{\sqrt{3}}{4}$ (2) $\frac{1}{2} + \frac{\sqrt{3}}{4}$ (3) $\frac{1}{2} - \frac{\sqrt{3}}{4}$ (4) $\frac{\sqrt{3}}{4} - \frac{1}{3}$

Ans. **[4]**

Sol. Co-ordinates of $P\left(\frac{1}{2}, 0\right)$, $Q\left(\frac{1}{2}, \frac{1}{2}\right)$, $R\left(\frac{\sqrt{3}}{2}, 1 - \frac{\sqrt{3}}{2}\right)$ and $S\left(\frac{\sqrt{3}}{2}, 0\right)$



Required area = Area of trapezium PQRS – $\int_{1/2}^{\sqrt{3}/2} \left(x - \frac{1}{2}\right)^2 dx$

$= \frac{1}{2}\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right)\left(\frac{1}{2} + 1 - \frac{\sqrt{3}}{2}\right) - \frac{1}{3}\left[\left(x - \frac{1}{2}\right)^3\right]_{1/2}^{\sqrt{3}/2}$

$= \frac{1}{2}\left(\frac{\sqrt{3}-1}{2}\right)\left(\frac{3-\sqrt{3}}{2}\right) - \frac{1}{3}\left[\left(\frac{\sqrt{3}-1}{2}\right)^3 - 0\right]$

$= \frac{\sqrt{3}}{4} - \frac{1}{3}$

Q.2 Let a, b $\in$ R, a $\neq$ 0 be such that the equation, $ax^2 - 2bx + 5 = 0$ has a repeated root α, which is also a root of the equation, $x^2 - 2bx - 10 = 0$. If β is the other root of this equation, then $\alpha^2 + \beta^2$ is equal to -

(1) 26 (2) 24 (3) 25 (4) 28

Ans. **[3]**

Sol. Roots of equation $ax^2 - 2bx + 5 = 0$ are α, α

$\therefore \alpha + \alpha = \frac{2b}{a} \Rightarrow \alpha = \frac{b}{a}$ ...(i)

and $\alpha^2 = \frac{5}{a}$ ...(ii)

from eq. (i) & (ii)

$\frac{b^2}{a^2} = \frac{5}{a} \Rightarrow b^2 = 5a$ ...(iii) $(a \neq 0)$

Roots of equation $x^2 - 2bx - 10 = 0$ are α, β

$\therefore \quad \alpha + \beta = 2b$

and $\quad \alpha\beta = -10$

Now, $\alpha = \frac{b}{a}$ is root of equation $x^2 - 2bx - 10 = 0$

$\therefore \frac{b^2}{a^2} - \frac{2b^2}{a} - 10 = 0$

$\Rightarrow \frac{5}{a} - 10 - 10 = 0 \qquad (\because b^2 = 5a)$

$\Rightarrow a = \frac{1}{4}$ and $b^2 = \frac{5}{4}$

So, $\alpha^2 = \frac{b^2}{a^2} = 20$ and $\beta^2 = 5$

$\Rightarrow \alpha^2 + \beta^2 = 25$

Q.3 If $A = \{x \in R : |x| < 2\}$ and $B = \{x \in R : |x - 2| \geq 3\}$; then -

(1) $A \cup B = R - (2, 5)$ (2) $A \cap B = (-2, -1)$ (3) $B - A = R - (-2, 5)$ (4) $A - B = [-1, 2)$

Ans. **[3]**

Sol. $A = \{x \in R : |x| < 2\}$

$A \in (-2, 2)$

and $B = \{x \in R : |x - 2| \geq 3\}$

$B \in (-\infty, -1] \cup [5, \infty)$

$A \cup B \in (-\infty, 2) \cup [5, \infty)$

$A \cap B \in (-2, -1]$

$B - A \in (-\infty, -2] \cup [5, \infty)$ or $R - (-2, 5)$

$A - B \in (-1, 2)$

Q.4 Let a_n be the nth term of a G.P. of positive terms. If $\sum_{n=1}^{100} a_{2n+1} = 200$ and $\sum_{n=1}^{100} a_{2n} = 100$, then $\sum_{n=1}^{200} a_n$ is equal to-

(1) 175 (2) 225 (3) 300 (4) 150

Ans. **[4]**

Sol. Let the G.P. is a, ar, ar^2 ....

$$\sum_{n=1}^{100} a_{2n+1} = 200 \Rightarrow a_3 + a_5 + a_7 + ... + a_{201} = 200$$

$$\Rightarrow \frac{ar^2(r^{200}-1)}{r^2-1} = 200 \quad ...(1)$$

$$\sum_{n=1}^{100} a_{2n} = 100 \Rightarrow a_2 + a_4 + ... + a_{200} = 100$$

$$\Rightarrow \frac{ar(r^{200}-1)}{r^2-1} = 100 \quad ...(2)$$

dividing (1) by (2)

we get, r = 2

adding both eq. (1) & (2)

$\Rightarrow a_2 + a_3 + a_4 + a_5 + ... + a_{201} = 300$

$\Rightarrow r(a_1 + a_2 + ... + a_{200}) = 300$

$$\Rightarrow a_1 + a_2 + ... + a_{200} = \frac{300}{r}$$

$$\Rightarrow \sum_{n=1}^{200} a_1 = \frac{300}{2} = 150$$

Q.5 If z be a complex number satisfying |Re(z)| + |Im(z)| = 4, then |z| cannot be -

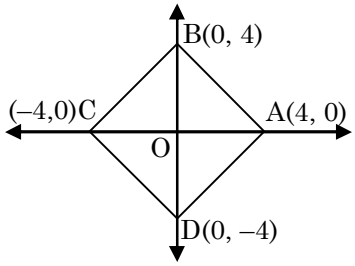
(1) $\sqrt{10}$ (2) $\sqrt{8}$ (3) $\sqrt{7}$ (4) $\sqrt{\frac{17}{2}}$

Ans. **[3]**

Sol. Let z = x + iy

given that |Re(z)| + |Im(z)| = 4

$\therefore$ |x| + |y| = 4



Maximum value of $|z| = 4$

Minimum value of $|z|$ = perpendicular distance of line AB from (0, 0)

$= 2\sqrt{2}$

So, $|z| \in [2\sqrt{2}, 4]$

So, $|z|$ can not be $\sqrt{7}$

Q.6 If $x = \sum_{n=0}^{\infty}(-1)^n \tan^{2n}\theta$ and $y = \sum_{n=0}^{\infty}\cos^{2n}\theta$, for $0 < \theta < \frac{\pi}{4}$, then -

(1) $y(1-x) = 1$ (2) $x(1-y) = 1$ (3) $y(1+x) = 1$ (4) $x(1+y) = 1$

Ans. **[1]**

Sol. $x = \sum_{n=0}^{\infty}(-1)^n \tan^{2n}\theta$

$\Rightarrow x = 1 - \tan^2\theta + \tan^4\theta - \tan^6\theta$ .....

$\Rightarrow x = \frac{1}{1+\tan^2\theta} = \cos^2\theta$ ...(1)

and $y = \sum_{n=0}^{\infty}\cos^{2n}\theta$

$y = 1 + \cos^2\theta + \cos^4\theta + \cos^6\theta + ...$

$y = \frac{1}{1-\cos^2\theta} = \frac{1}{\sin^2\theta}$

$\Rightarrow \sin^2\theta = \frac{1}{y}$ ....(2)

Adding (1) & (2), we get,

$x + \frac{1}{y} = 1$

$\Rightarrow \quad y(1-x) = 1$

Q.7 Let [t] denote the greatest integer $\le$ t and $\lim_{x\to 0} x\left[\frac{4}{x}\right] = A$. Then the function, $f(x) = [x^2]\sin(\pi x)$ is discontinuous, when x is equal to -

(1) $\sqrt{A+1}$ (2) $\sqrt{A}$ (3) $\sqrt{A+21}$ (4) $\sqrt{A+5}$

Ans. **[1]**

Sol. $\lim_{x\to 0} x\left[\frac{4}{x}\right] = A$

$\Rightarrow \lim_{x\to 0} x\left(\frac{4}{x} - \left\{\frac{4}{x}\right\}\right) = A$

$\Rightarrow \lim_{x\to 0}\left(4 - x\left\{\frac{4}{x}\right\}\right) = A$

$\Rightarrow A = 4$

Now, $f(x) = [x^2]\sin(\pi x)$ is continuous at every integer point but discontinuous at non integer points then by options, $\sqrt{A+1}$ is correct answer.

Q.8 If $p \rightarrow (p \wedge \sim q)$ is false, then the truth values of p and q are respectively -

(1) F, F (2) T, F (3) T, T (4) F, T

Ans. **[3]**

Sol. $p \rightarrow (p \wedge \sim q)$ will be false only when p is true and $(p \wedge \sim q)$ is false.

So, $p = T, q = T$

Q.9 Let a function $f : [0, 5] \rightarrow R$ be continuous, $f(1) = 3$ and F be defined as $F(x) = \int_1^x t^2 g(t)\,dt$, where $g(t) = \int_1^t f(u)\,du$. Then for the function F, the point x = 1 is

(1) not a critical point (2) a point of local maxima

(3) a point of local minima (4) a point of inflection

Ans. **[3]**

Sol. $F(x) = \int_1^x t^2 g(t)\,dt$

$\Rightarrow F'(x) = x^2 g(x)$

at $F'(1) = (1)\, g(1) = 0 \quad \{\because g(1) = 0\}$

Now, $F''(x) = 2xg(x) + x^2 g'(x)$

$F''(1) = 2g(1) + g'(1)$

$F''(1) = 0 + g'(1) \quad \{g'(t) = f(t);\ g'(1) = f'(1) = 3\}$

$F''(1) = 3$

So, at $x = 1$, $F'(1) = 0$ but $F''(1) > 0$

$\therefore$ For the function f(x), x = 1 is a point of local minima.

Q.10 A random variable X has the following probability distribution :

X	:	1	2	3	4	5
P(X)	:	K^2	2K	K	2K	$5K^2$

Then P(X > 2) is equal to -

(1) $\frac{1}{36}$ (2) $\frac{1}{6}$ (3) $\frac{7}{12}$ (4) $\frac{23}{36}$

Ans. **[4]**

Sol. $\sum_{i=1}^{5} P(X) = 1$

$\Rightarrow K^2 + 2K + K + 2K + 5K^2 = 1$

$\Rightarrow 6K^2 + 5K - 1 = 0$

$\Rightarrow K = \frac{1}{6},\ K = -1$ (rejected)

So, $K = \frac{1}{6}$

$P(X > 2) = K + 2K + 5K^2$

$= \frac{1}{6} + \frac{2}{6} + \frac{5}{36}$

$= \frac{23}{36}$

Q.11 If $\int \frac{d\theta}{\cos^2\theta(\tan 2\theta + \sec 2\theta)} = \lambda \tan\theta + 2\log_e |f(\theta)| + C$ where C is a constant of integration, then the ordered pair $(\lambda, f(\theta))$ is equal to -

(1) $(1, 1 - \tan\theta)$ (2) $(-1, 1 - \tan\theta)$ (3) $(-1, 1 + \tan\theta)$ (4) $(1, 1 + \tan\theta)$

Ans. **[3]**

Sol. $\int \frac{d\theta}{\cos^2\theta(\tan 2\theta + \sec 2\theta)}$

$= \int \frac{d\theta}{\cos^2\theta\left(\frac{2\tan\theta}{1-\tan^2\theta} + \frac{1+\tan^2\theta}{1-\tan^2\theta}\right)}$

$= \int \frac{\sec^2\theta\,(1-\tan^2\theta)}{(1+\tan\theta)^2}\,d\theta$

Put $\tan\theta = t$

$\Rightarrow \sec^2\theta\, d\theta = dt$

$= \int \frac{(1-t^2)}{(1+t)^2}\,dt = \int\left(\frac{1-t}{1+t}\right)dt$

$= \int\left(-1 + \frac{2}{1+t}\right)dt$

$= -t + 2\log_e|1 + t| + C$

$= -\tan\theta + 2\log_e|1 + \tan\theta| + C$

$\therefore\ \lambda = -1$ and $f(\theta) = 1 + \tan\theta$

Q.12 The length of the minor axis (along y-axis) of an ellipse in the standard form is $\frac{4}{\sqrt{3}}$. If this ellipse touches the line, $x + 6y = 8$, then its eccentricity is -

(1) $\frac{1}{2}\sqrt{\frac{11}{3}}$ (2) $\frac{1}{3}\sqrt{\frac{11}{3}}$ (3) $\frac{1}{2}\sqrt{\frac{5}{3}}$ (4) $\sqrt{\frac{5}{6}}$

Ans. **[1]**

Sol. Let the equation of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,\ (a > b)$

Given that $2b = \frac{4}{\sqrt{3}} \Rightarrow b = \frac{2}{\sqrt{3}}$

Equation of tangent $y = mx \pm \sqrt{a^2m^2 + b^2}$ ....(1)

Given tangent is $x + 6y = 8$

$$\Rightarrow \quad y = -\frac{1}{6}x + \frac{8}{6} \qquad ...(2)$$

from eq. (1) & (2)

$$m = -\frac{1}{6} \text{ and } a^2m^2 + b^2 = \frac{16}{9}$$

$$\Rightarrow \quad a^2\left(\frac{1}{36}\right) + \frac{4}{3} = \frac{16}{9}$$

$$\Rightarrow \quad a^2 = 16$$

Now, $e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{4/3}{16}} = \sqrt{\frac{11}{12}} = \frac{1}{2}\sqrt{\frac{11}{3}}$

Q.13 In the expansion of $\left(\frac{x}{\cos\theta} + \frac{1}{x\sin\theta}\right)^{16}$, if ℓ_1 is the least value of the term independent of x when $\frac{\pi}{8} \le \theta \le \frac{\pi}{4}$ and ℓ_2 is the least value of the term independent of x when $\frac{\pi}{16} \le \theta \le \frac{\pi}{8}$, then the ratio $\ell_2 : \ell_1$ is equal to -

(1) 16 : 1 (2) 1 : 8 (3) 8 : 1 (4) 1 : 16

Ans. **[1]**

Sol. General term $T_{r+1} = {}^{16}C_r\left(\frac{x}{\cos\theta}\right)^{16-r}\left(\frac{1}{x\sin\theta}\right)^r$

$$T_{r+1} = {}^{16}C_r\left(\frac{1}{\cos\theta}\right)^{16-r}\left(\frac{1}{\sin\theta}\right)^r x^{16-2r}$$

term is independent of x

$\therefore$ $16 - 2r = 0$

$\Rightarrow r = 8$

$$T_9 = {}^{16}C_8\left(\frac{1}{\cos\theta}\right)^8\left(\frac{1}{\sin\theta}\right)^8$$

$$T_9 = {}^{16}C_8\frac{2^8}{(\sin 2\theta)^8}$$

$$\theta \in \left[\frac{\pi}{8}, \frac{\pi}{4}\right] \Rightarrow 2\theta \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$$

For least value $\Rightarrow \sin2\theta$ should be maximum

$$\therefore \ 2\theta = \frac{\pi}{2}$$

So, $\ell_1 = {}^{16}C_8(2^8)$ ...(1)

Again, $\theta \in \left[\frac{\pi}{16}, \frac{\pi}{8}\right] \Rightarrow 2\theta \in \left[\frac{\pi}{8}, \frac{\pi}{4}\right]$

Minimum value, $\ell_2 = {}^{16}C_8 \dfrac{2^8}{(1/\sqrt{2})^8} = {}^{16}C_8 2^{12}$ ...(2)

$\dfrac{\ell_2}{\ell_1} = \dfrac{2^4}{1} = \dfrac{16}{1}$

Q.14 If $x = 2\sin\theta - \sin2\theta$ and $y = 2\cos\theta - \cos2\theta$, $\theta \in [0, 2\pi]$, then $\dfrac{d^2y}{dx^2}$ at $\theta = \pi$ is -

(1) $\dfrac{3}{2}$ (2) $\dfrac{3}{4}$ (3) $-\dfrac{3}{4}$ (4) $-\dfrac{3}{8}$

Ans. **[Bonus]**

Sol. $x = 2\sin\theta - \sin2\theta$

$\Rightarrow \dfrac{dx}{d\theta} = 2\cos\theta - 2\cos2\theta$

and $y = 2\cos\theta - \cos 2\theta$

$\Rightarrow \dfrac{dy}{d\theta} = -2\sin\theta + 2\sin2\theta$

Now, $\dfrac{dy}{dx} = \dfrac{2(\sin 2\theta - \sin\theta)}{(\cos\theta - \cos 2\theta)} = \dfrac{2\cos\dfrac{3\theta}{2}\sin\dfrac{\theta}{2}}{2\sin\dfrac{3\theta}{2}.\sin\dfrac{\theta}{2}}$

$\dfrac{dy}{dx} = \cot\dfrac{3\theta}{2}$

$\Rightarrow \dfrac{d^2y}{dx^2} = -\operatorname{cosec}^2\dfrac{3\theta}{2}\left(\dfrac{3}{2}\right)\dfrac{d\theta}{dx}$

$\Rightarrow \dfrac{d^2y}{dx^2} = -\dfrac{3}{2}\left(\operatorname{cosec}^2\dfrac{3\theta}{2}\right)\left(\dfrac{1}{2\cos\theta - 2\cos 2\theta}\right)$

at $\theta = \pi$

$\dfrac{d^2y}{dx^2} = \left(-\dfrac{3}{2}\right)(1)\left(\dfrac{1}{-2-2}\right) = \dfrac{3}{8}$

Q.15 If $\dfrac{dy}{dx} = \dfrac{xy}{x^2+y^2}$; $y(1) = 1$; then a value of x satisfying $y(x) = e$ is -

(1) $\dfrac{1}{2}\sqrt{3}\,e$ (2) $\sqrt{2}\,e$ (3) $\dfrac{e}{\sqrt{2}}$ (4) $\sqrt{3}\,e$

Ans. **[4]**

Sol. $\dfrac{dy}{dx} = \dfrac{xy}{x^2+y^2}$

Put $y = vx$

then $\dfrac{dy}{dx} = v + x\dfrac{dv}{dx}$

$\Rightarrow v + x\frac{dv}{dx} = \frac{x(vx)}{x^2 + v^2x^2}$

$\Rightarrow x\frac{dv}{dx} = \frac{v}{1+v^2} - v$

$\Rightarrow \int \frac{1+v^2}{v^3}\,dv = -\int \frac{dx}{x}$

$\Rightarrow -\frac{1}{2v^2} + \log v = -\log x + C$

$\Rightarrow -\frac{1}{2}\frac{x^2}{y^2} + \log\left(\frac{y}{x}\right) = -\log x + C$ ...(1)

put $x = 1, y = 1$

$\Rightarrow C = -\frac{1}{2}$

from eq. (1)

$-\frac{1}{2}\frac{x^2}{y^2} + \log\left(\frac{y}{x}\right) = -\log x - \frac{1}{2}$

Put $y = e \Rightarrow -\frac{x^2}{2e^2} + \log\left(\frac{e}{x}\right) + \log x = -\frac{1}{2}$

$\Rightarrow x^2 = 3e^2$

$\Rightarrow x = \pm\sqrt{3}\,e$

$\Rightarrow x = \sqrt{3}\,e$

Q.16 Let $a - 2b + c = 1$. If $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$, then -

(1) $f(-50) = -1$ (2) $f(50) = 1$ (3) $f(50) = -501$ (4) $f(-50) = 501$

Ans. **[2]**

Sol. $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$

$R_1 \to R_1 - 2R_2 + R_3$

$f(x) = \begin{vmatrix} 1 & 0 & 0 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$

$f(x) = 1$

$\therefore\ f(50) = 1$

Q.17 If 10 different balls are to be placed in 4 distinct boxes at random, then the probability that two of these boxes contain exactly 2 and 3 balls is -

(1) $\frac{945}{2^{10}}$ (2) $\frac{965}{2^{11}}$ (3) $\frac{945}{2^{11}}$ (4) $\frac{965}{2^{10}}$

Ans. **[1]**

Sol. Total ways $\Rightarrow n = 4^{10}$

Number of ways placing exactly 2 and 3 balls in two of these boxes

$= {}^4C_2 \times \frac{5!}{2!\ 3!} \times 2! \times {}^{10}C_5 \times 2^5$

Required probability $= \frac{945}{2^{10}}$

Q.18 The following system of linear equations

$7x + 6y - 2z = 0$

$3x + 4y + 2z = 0$

$x - 2y - 6z = 0$, has

(1) no solution

(2) infinitely many solutions, (x, y, z) satisfying $y = 2z$

(3) only the trivial solution

(4) infinitely many solutions, (x, y, z) satisfying $x = 2z$

Ans. **[4]**

Sol. $7x + 6y - 2z = 0$ ....(1)

$3x + 4y + 2z = 0$ ....(2)

$x - 2y - 6z = 0$ ....(3)

$$\Delta = \begin{vmatrix} 7 & 6 & -2 \\ 3 & 4 & 2 \\ 1 & -2 & -6 \end{vmatrix}$$

$\Delta = 7(-24 + 4) - 6(-18 - 2) - 2(-6 - 4) = 0$

$\Delta = 0$

$\therefore$ infinite non-trival solution exist

to eliminate y we operate eq. (1) – (2) + (3)

$5x = 10z$

$x = 2z$

Q.19 If one end of a focal chord AB of the parabola $y^2 = 8x$ is at $A\left(\frac{1}{2}, -2\right)$, then the equation of the tangent to it at B is -

(1) $2x + y - 24 = 0$ (2) $x + 2y + 8 = 0$ (3) $x - 2y + 8 = 0$ (4) $2x - y - 24 = 0$

Ans. **[3]**

Sol. $y^2 = 8x$ $(a = 2)$

Let one end of focal chord is $(at^2, 2at) = \left(\frac{1}{2}, -2\right)$

$\Rightarrow 2at = -2$

$\Rightarrow t = -1/2$

other end of focal chord will be $\left(\frac{a}{t^2}, -\frac{2a}{t}\right) \equiv (8, 8)$

Now, tangent at B(8, 8)

$\Rightarrow y(8) = 8\left(\frac{x+8}{2}\right)$

$\Rightarrow x - 2y + 8 = 0$

Q.20 Let f and g be differentiable functions on R such that fog is the identity function. If for some a, b $\in$ R, $g'(a) = 5$ and $g(a) = b$, then $f'(b)$ is equal to -

(1) $\frac{1}{5}$ (2) $\frac{2}{5}$ (3) 5 (4) 1

Ans. **[1]**

Sol. fog is an identity function

$\therefore$ $fog(x) = x$

$\Rightarrow f'\{g(x)\}\, g'(x) = 1$

put $x = a$

$\Rightarrow f'\{g(a)\}\, g'(a) = 1$

$\Rightarrow f'(b)\,(5) = 1$

$\Rightarrow f'(b) = \frac{1}{5}$

Q.21 If the distance between the plane, $23x - 10y - 2z + 48 = 0$ and the plane containing the lines $\frac{x+1}{2} = \frac{y-3}{4} = \frac{z+1}{3}$ and $\frac{x+3}{2} = \frac{y+2}{6} = \frac{z-1}{\lambda}$ ($\lambda \in$ R) is equal to $\frac{k}{\sqrt{633}}$, then k is equal to _____.

Ans. **[3]**

Sol. Required distance = perpendicular distance of plane $23x - 10y - 2z + 48 = 0$ either from (–1, 3, –1) or (–3, –2, 1)

$\Rightarrow \left|\frac{-23-30+2+48}{\sqrt{(23)^2+(10)^2+(2)^2}}\right| = \frac{k}{\sqrt{633}}$

$\Rightarrow \frac{3}{\sqrt{633}} = \frac{k}{\sqrt{633}}$

$\therefore$ $k = 3$

Q.22 If $C_r = {}^{25}C_r$ and $C_0 + 5\cdot C_1 + 9\cdot C_2 + \ldots. + (101)\cdot C_{25} = 2^{25}\cdot k$, then k is equal to _____.

Ans. **[51]**

Sol. $C_0 + 5\cdot C_1 + 9\cdot C_2 + \ldots. + (101)\cdot C_{25}$

$= \sum_{r=0}^{25}(4r+1)\,{}^{25}C_r = 4\sum_{r=0}^{25} r.\,{}^{25}C_r + \sum_{r=0}^{25}{}^{25}C_r$

$= 4\sum_{r=0}^{25} r.\frac{25}{r}{}^{24}C_{r-1} + 2^{25}$

$= 100 \cdot 2^{24} + 2^{25}$

$= 2^{25}(50 + 1)$

$\therefore k = 51$

Q.23 The number of terms common to the two A.P.'s 3, 7, 11, ….. 407 and 2, 9, 16, …. 709 is _____.

Ans. **[14]**

Sol. First A.P. is 3, 7, 11, 15, 19, 23, ... 407

Second A.P. is 2, 9, 16, 23, ... 709

First common term = 23

Common difference d = L.C.M. (4, 7) = 28

Last term $\le$ 407

$\Rightarrow 23 + (n - 1)(28) \le 407$

$\Rightarrow n \le 14.7$

So, n = 14

Q.24 If the curves, $x^2 - 6x + y^2 + 8 = 0$ and $x^2 - 8y + y^2 + 16 - k = 0$, $(k > 0)$ touch each other at a point, then the largest value of k is _____.

Ans. **[36]**

Sol. $C_1 = (3, 0)$, $C_2(0, 4)$

$r_1 = \sqrt{9+0-8} = 1$; $r_2 = \sqrt{16-16+k} = \sqrt{k}$

Two circles touch each other

$\therefore C_1C_2 = |r_1 \pm r_2|$

$5 = |1 \pm \sqrt{k}|$

$\Rightarrow 1 + \sqrt{k} = 5$ or $\sqrt{k} - 1 = 5$

$\Rightarrow k = 16$ or $k = 36$

Maximum value of k = 36

Q.25 Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors such that $|\vec{a}| = \sqrt{3}$, $|\vec{b}| = 5$, $\vec{b} \cdot \vec{c} = 10$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{3}$. If $\vec{a}$ is perpendicular to the vector $\vec{b} \times \vec{c}$, then $|\vec{a} \times (\vec{b} \times \vec{c})|$ is equal to _____.

Ans. **[30]**

Sol. $|\vec{a} \times (\vec{b} \times \vec{c})| = |\vec{a}||\vec{b} \times \vec{c})| \sin\frac{\pi}{2}$

$= |\vec{a}||\vec{b}||\vec{c}| \sin\frac{\pi}{3}$

$$= (\sqrt{3})\,(5)\,|\vec{c}\,|.\frac{\sqrt{3}}{2}$$

$$= \frac{15}{2}|\vec{c}\,| \qquad ...(1)$$

$$\Rightarrow \cos\frac{\pi}{3} = \frac{\vec{b}\,.\,\vec{c}}{|\vec{b}\,||\vec{c}\,|}$$

$$\Rightarrow \frac{1}{2} = \frac{10}{5|\vec{c}\,|} \Rightarrow |\vec{c}\,| = 4$$

from eq. (1)

$$|\vec{a}\times(\vec{b}\times\vec{c})| = \frac{15}{2}\times 4 = 30$$